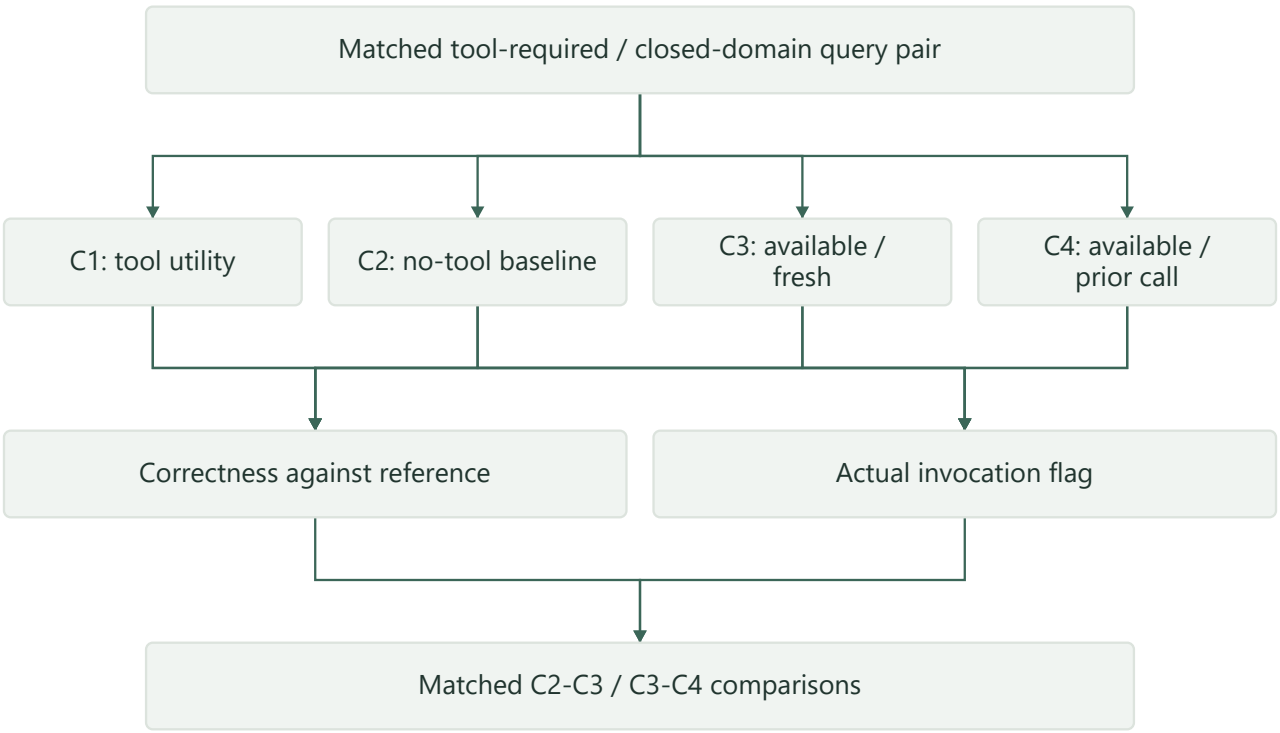
When Tools Get in the Way

Tool-use research · Paper 2

A paired experiment testing whether a related but unnecessary tool changes answers to questions answerable without it, separating availability, prior interaction, correctness and actual invocation.

Architecture



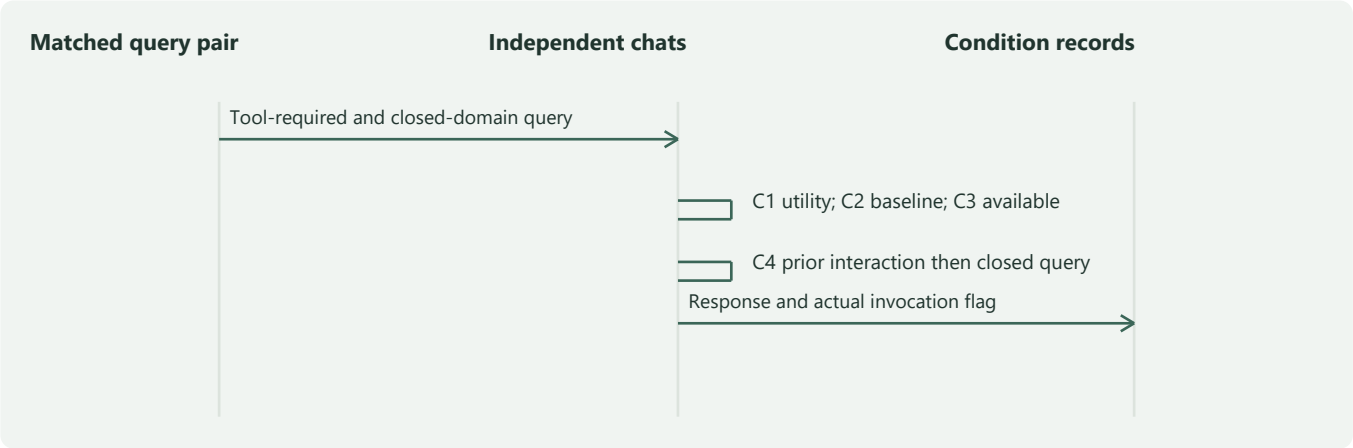
Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Pair construction	Tool-required + closed-domain	Topically adjacent, different scope
Conditions	C1 / C2 / C3 / C4	Utility, baseline, availability, history
Measurement	Correctness + invocation	Separate outcome variables
Controls	Same closed query in C2-C4	Within-pair comparison

Research preprint

Paired queries and four conditions

Paper 2 studies a different intervention: a related tool is available even though the closed-domain question can be answered from model knowledge. Each domain pairs a tool-required query with a nearby closed-domain query and a checkable answer key. The three FULL notebooks each contain fifty such pairs.



The controlled comparison

C1 asks the tool-required query with a tool in a fresh chat. C2 asks the closed-domain query without a tool. C3 asks that identical closed-domain query with the tool available. C4 repeats it after a preceding tool-domain interaction in the same chat. The notebook creates separate chats for all branches; only C4 intentionally carries history.

Tool scope

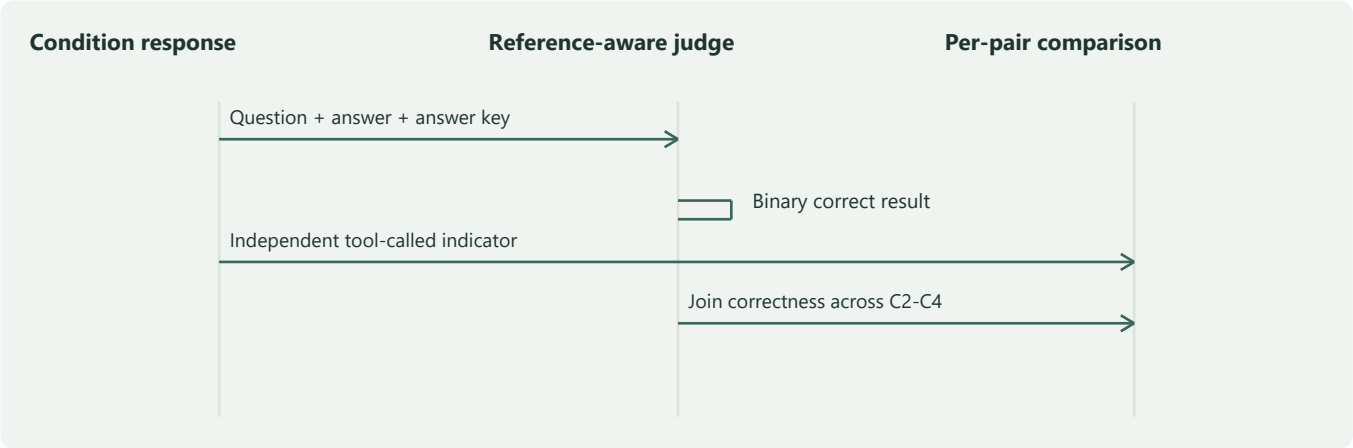
The domain lookup returns a stored answer keyed by the current paired tool query. It is not a general-purpose knowledge tool. An unnecessary invocation can therefore return the paired record rather than the closed-domain answer. Keeping query scope visible explains why tool availability and tool invocation are different variables.

Condition	Tool / history	Query
C1 utility control	Available / fresh	Tool-required
C2 vs C3	Absent vs available / fresh	Same closed-domain query
C3 vs C4	Available / fresh vs primed	Same closed-domain query

Source / interpretation: paper2_geography_FULL_(2), paper2_biology_FULL_(2), paper2_astronomy_FULL_(3): cells 10,12,14,16,20. Paper: arXiv:2609.14157v1, section 3.

Answering and invocation measures

The source records answer correctness separately from whether the tool was invoked. This lets the analysis detect a model that avoids an unnecessary call yet still withholds a correct answer. A call count alone would miss that behavior.



Correctness scoring

The judge receives the question, reference answer and final response. It returns a JSON boolean at temperature zero, ignoring wording and formatting differences when the final answer is correct. The notebook computes answer rate as mean correctness times 100 and tool-use rate as mean actual-call flags times 100. These measures must remain separate in charts and interpretation.

History attribution

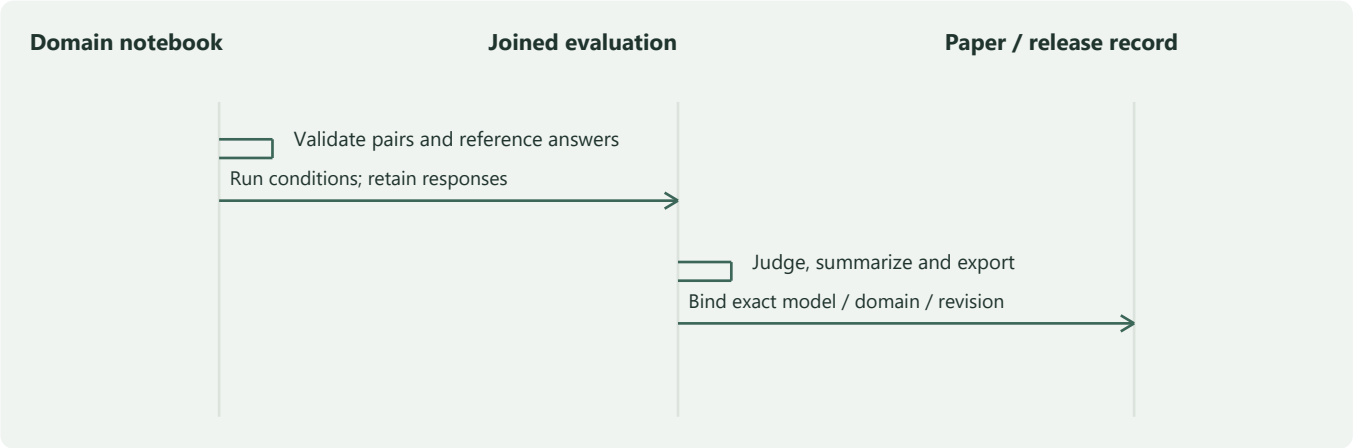
In C4 the code sends the tool-domain prompt first, resets the invocation flag, then sends the closed-domain query. The resulting tool-use flag measures the second query rather than merely counting the priming interaction. The harness does not force priming to invoke a tool, so final evidence should verify the preceding call and its success before treating every row as successfully primed.

Analysis	What it distinguishes
C2 to C3 correctness	Effect of unnecessary availability
C3 to C4 correctness	Effect of prior interaction
Tool-use versus correctness	Invocation choice versus answering behavior

Source / interpretation: All three paper2 FULL notebooks: cell 20 for execution/reset, cells 24 and 26-31 for invocation, judge and aggregation. No new model evaluation was executed.

Workflow evidence and release lineage

The supplied notebooks document currency-pilot and geography, biology and astronomy workflows. The final paper reports a larger ten-domain, six-model study. A technical portfolio should show how the supplied workflows implement the design without presenting this subset as a complete reconstruction of every published run.



From pilot to larger study

The currency pilot uses thirty trials, while the three FULL notebooks assert fifty pairs per domain. Each FULL notebook has one configured historical model. Their source establishes treatment construction and measurement, but does not itself supply all ten domains, six models or the scope-aware mitigation runs. Final totals and reported findings should link to the paper and corresponding run archive.

A clean reproducibility package

Keep a safe paired-query dataset, tool-scope definitions, four-condition runner, reference-aware scorer and aggregation scripts. Record model identifier, generation settings, SDK behavior, pair identity and errors with each run. Spreadsheet export is a reporting operation rather than part of the treatment. The original notebooks contain credential literals, so a public release should use external credentials and cleared outputs rather than copying the files wholesale.

Artifact	Role
Pair and tool definitions	Define the matched question and scoped utility
Condition and response records	Trace availability/history and actual calls
Scorer and joined table	Reproduce answer-rate and trajectory analyses

Source / interpretation: Copy_of_workflowreadonly (1), cells 5,7,13,17-19; all paper2 FULL notebooks, cells 16,20,26-33. Paper-reported full-study scope: arXiv:2609.14157v1.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Paper scope	500 pairs; ten domains; six models	arXiv:2609.14157
Supplied workflow	Three 50-pair domain notebooks	Geography, biology, astronomy
Earlier pilot	30 currency trials	Copy_of_workflowreadonly

Primary references

File / record	Path / reference
arXiv manuscript · Paper 2	documents/tool-failure-paper-2-manuscript.pdf
Read the paper on arXiv	https://arxiv.org/abs/2609.14157

The technical companion reconstructs the experiment workflow from the supplied notebook code and links the paper. Raw notebooks are not distributed here; the full repository release will use cleaned reusable code and versioned run records.

Evidence boundary

Study findings are attributed to the linked preprint. Notebook inspection establishes experiment mechanisms and controls; it does not constitute a fresh reproduction of the full study.

Technical note

Paper 1 studies post-failure reporting; Paper 2 studies unnecessary-tool availability and history. Their treatments and outcome measures are kept distinct.